

Name:	ID:	Section:
-------	-----	----------

Question 1 [C01] [10 Points]

The water park needs a program to manage visitor eligibility for various activities and special offers. Write a Java program that first takes the number of visitor sessions, followed by the number of visitors in each session. For each visitor, the program should further take the Activity Level (1 for Low, 2 for Moderate, 3 for High) and whether the visitor has visited the park previously (true for previous visitors, false for first-time visitors). The park offers the following activities and special offers: the **Lazy Pool** is available for visitors with an activity level of 2 or 3; the **Water Coaster** is available for visitors with an activity level of 3 who have visited the park before; and the **Special Offer** is available to first-time visitors. The program should count and display the total number of eligible visitors for each activity and for the Special Offer.

Sample Input	Sample Output
Enter number of visitor sessions: 2 Enter number of visitors in session 1: 3 Enter activity level of visitor 1: 2 Enter previous visit of visitor 1: true Enter activity level of visitor 2: 1 Enter previous visit of visitor 2: false Enter activity level of visitor 3: 2 Enter previous visit of visitor 3: false Enter number of visitors in session 2: 2 Enter activity level of visitor 1: 3 Enter previous visit of visitor 1: true Enter activity level of visitor 2: 2 Enter previous visit of visitor 2: false	Lazy Pool Riders: 4 Water Coaster Riders: 1 Visitors with Special Offer: 3 Explanation: For session 1, Visitor 1 qualifies for the Lazy Pool but not the Water Coaster due to an activity level of 2. Visitor 2 qualifies for the Special Offer as a first-time visitor. Visitor 3 qualifies for the Lazy Pool and Special Offer. For session 2, Visitor 1 qualifies for both the Lazy Pool and the Water Coaster. Visitor 2 qualifies for the Lazy Pool and the Special Offer.

Name:	ID:	Section:
-------	-----	----------

Question 1 [C01] [10 Points]

You are asked to write a Java program for an amusement park that manages visitor eligibility for multiple rides and special offers. The program first takes the number of entry sessions from the management team. For each session, the program takes the number of visitors, followed by each visitor's height (in cm) and age (in years). The park has multiple rides with eligibility rules: the **Roller Coaster** requires a minimum height of 140 cm and a minimum age of 12 years, the **Ferris Wheel** allows visitors aged 5 years or older, and the **Bumper Cars** allow visitors aged 8 years or older with a maximum height of 200 cm for safety. Additionally, visitors under 12 years are eligible for a **Special Discount Offer**. The program should count the number of visitors who are eligible for each ride, as well as how many receive the discount. Finally, the program should display these counts after processing all sessions.

Sample Input	Sample Output
Enter number of sessions = 3 Enter number of visitors in session 1 = 3 Enter height of visitor 1: 150 Enter age of visitor 1: 14 Enter height of visitor 2: 120 Enter age of visitor 2: 10 Enter height of visitor 3: 160 Enter age of visitor 3: 20 Enter number of visitors in session 2 = 2 Enter height of visitor 1: 130 Enter age of visitor 1: 6 Enter height of visitor 2: 170 Enter age of visitor 2: 13 Enter number of visitors in session 3 = 1 Enter height of visitor 1: 180 Enter age of visitor 1: 9	Roller Coaster Riders: 3 Ferris Wheel Riders: 6 Bumper Cars Riders: 5 Visitors with Special Offer: 3 Explanation: Visitor 1 in session 1 qualifies for all rides, Visitor 2 qualifies for the Ferris Wheel and Bumper Cars and gets the Special Offer, and Visitor 3 qualifies for all rides. In session 2, Visitor 1 qualifies for the Ferris Wheel and the Special Offer, while Visitor 2 qualifies for all rides. In session 3, the visitor qualifies for the Ferris Wheel and Bumper Cars and receives the Special Offer.

In case of any circumstance not present in the rubrics, feel free to discuss in the assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

SET - [A]

SOLUTION

```
import java.util.Scanner;

public class WaterPark {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int lazyPool = 0, waterCoaster = 0, specialOffer = 0;

        System.out.print("Enter number of visitor sessions: ");
        int sessions = sc.nextInt();

        for (int i = 1; i <= sessions; i++) {
            System.out.print("Enter number of visitors in session " + i + ": ");
            int visitors = sc.nextInt();

            for (int j = 1; j <= visitors; j++) {
                System.out.print("Enter activity level of visitor " + j + ": ");
                int activityLevel = sc.nextInt();
                System.out.print("Enter previous visit of visitor " + j + ": ");
                boolean previousVisit = sc.nextBoolean();

                if (activityLevel == 2 || activityLevel == 3) {
                    lazyPool++;
                }

                if (activityLevel == 3 && previousVisit==true) {
                    waterCoaster++;
                }

                if (previousVisit==false) {
                    specialOffer++;
                }
            }
        }

        System.out.println("Lazy Pool Riders: " + lazyPool);
        System.out.println("Water Coaster Riders: " + waterCoaster);
        System.out.println("Visitors with Special Offer: " + specialOffer);
    }
}
```

RUBRIC

SL	Points to Meet	Marks [10]
1	The program takes input for sessions, number of visitors, activity level, and previous visit	1
2	Correctly calculates eligibility for all activities (Lazy Pool, Water Coaster)	3
3	Correctly calculates eligibility for Special Offer	1
4	Displays total counts for all activities and Special Offers correctly	2
5	Proper use of nested loops and conditionals	3

SET - [B]**SOLUTION**

```
import java.util.Scanner;

public class AmusementPark {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int rollerCoaster = 0, ferrisWheel = 0, bumperCars = 0, specialOffer = 0;

        System.out.print("Enter number of sessions = ");
        int sessions = sc.nextInt();

        for (int i = 1; i <= sessions; i++) {
            System.out.print("Enter number of visitors in session " + i + "= ");
            int visitors = sc.nextInt();

            for (int j = 1; j <= visitors; j++) {
                System.out.print("Enter height of visitor " + j + ": ");
                int height = sc.nextInt();
                System.out.print("Enter age of visitor " + j + ": ");
                int age = sc.nextInt();

                if (height >= 140 && age >= 12) {
                    rollerCoaster++;
                }

                if (age >= 5) {
                    ferrisWheel++;
                }

                if (age >= 8 && height <= 200) {
                    bumperCars++;
                }

                if (age < 12) {
                    specialOffer++;
                }
            }
        }

        System.out.println("Roller Coaster Riders: " + rollerCoaster);
        System.out.println("Ferris Wheel Riders: " + ferrisWheel);
        System.out.println("Bumper Cars Riders: " + bumperCars);
        System.out.println("Visitors with Special Offer: " + specialOffer);
    }
}
```

RUBRIC		
SL	Points to Meet	Marks [10]
1	The program takes input for sessions, number of visitors, height, and age	1
2	Correctly calculates eligibility for all rides (Roller Coaster, Ferris Wheel, Bumper Cars)	3
3	Correctly calculates Special Offer eligibility	1
4	Displays total counts for all rides and Special Offers correctly	2
5	Proper use of nested loops	3

Name:	ID:	Section:
-------	-----	----------

Question 1 [C02] [10 Points]

Write a Java program that will take two Strings of the same length and an operator character (* or /) from the user. Your task is to create two numbers from the two input strings by considering only the odd digits and perform mathematical operations based on the third input (operator). Finally, print the newly created two numbers and the output value. The input String can contain other characters than digits. For simplicity, assume that both input strings will always be of the same length. You are not allowed to use any built-in String methods other than .length() and .charAt() methods.

Sample Input	Sample Output
ab4s3F67h%21 mnP543&1b80Q *	371 531 197001
5dQ!87 26f0&3 /	57 3 19
Explanation: For sample Input 1, ab4s3F67h%21 contains odd digits 3,7,1, that's why the first number is 371 mnP543&1b86Q contains odd digits 5,3,1, that's why the second number is 531 After performing the addition operation (*), the final output is 197001	

Name:	ID:	Section:
-------	-----	----------

Question 1 [C02] [10 Points]

Write a Java program that will take two Strings of the same length and an operator character (+ or -) from the user. Your task is to create two numbers from the two input strings by considering only the even digits and perform mathematical operations based on the third input (operator). Finally, print the newly created two numbers and the output value. The input String can contain other characters than digits. For simplicity, assume that both input strings will always be of the same length. You are not allowed to use any built-in String methods other than .length() and .charAt() methods.

Sample Input	Sample Output
ab4s3F67h%21 mnP543&1b80Q +	462 480 942
2dQ!83 36f1&5 -	28 6 22
Explanation: For sample Input 1, ab4s3F67h%21 contains even digits 4,6,2, that's why the first number is 462 mnP543&1b86Q contains even digits 4,8,0, that's why the second number is 480 After performing the addition operation (+), the final output is 942	

In case of any circumstance not present in the rubrics, feel free to discuss in the assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

SET - [A]

SOLUTION

```
import java.util.Scanner;
public class A{
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        String s1 = sc.next();
        String s2 = sc.next();
        char op = sc.next().charAt(0);
        int num1 = 0, num2 = 0;
        for(int i = 0; i<s1.length(); i++){
            char ch1 = s1.charAt(i);
            char ch2 = s2.charAt(i);
            if(ch1>='0' && ch1<='9'){
                int digit = (int)ch1 - 48;
                if(digit%2 != 0){
                    num1 = num1*10 + digit;
                }
            }
            if(ch2>='0' && ch2<='9'){
                int digit = (int)ch2 - 48;
                if(digit%2 != 0){
                    num2 = num2*10 + digit;
                }
            }
        }
        System.out.println(num1);
        System.out.println(num2);
        if(op == '*'){
            System.out.println(num1*num2);
        }
        else if(op == '/'){
            System.out.println(num1/num2);
        }
    }
}
```

RUBRIC

SL	Points to Meet	Marks [10]
1	Importing Scanner class, main method, and Java syntax	0.5
2	Correctly takes two strings and an operator from the user	1

3	Properly handles the iteration and extracts the characters of both Strings	1.5
4	Checking the digits using characters ('0'-'9') or ASCII values [both Strings]	1
5	Converting the character to a digit for both Strings	1
6	Searching the odd digits [both Strings]	0.5
7	Initialize num1, num2, and correctly prepare the numbers	3
8	Checking the operations (*,/) and performing the correct operation	1
9	Printing all three outputs correctly	0.5

SET - [B]

SOLUTION

```
import java.util.Scanner;
public class B{
    public static void main(String [] args){
        Scanner sc = new Scanner(System.in);
        String s1 = sc.next();
        String s2 = sc.next();
        char op = sc.next().charAt(0);
        int num1 = 0, num2 = 0;
        for(int i = 0; i<s1.length(); i++){
            char ch1 = s1.charAt(i);
            char ch2 = s2.charAt(i);
            if(ch1>='0' && ch1<='9'){
                int digit = (int)ch1 - 48;
                if(digit%2 == 0){
                    num1 = num1*10 + digit;
                }
            }
            if(ch2>='0' && ch2<='9'){
                int digit = (int)ch2 - 48;
                if(digit%2 == 0){
                    num2 = num2*10 + digit;
                }
            }
        }
        System.out.println(num1);
        System.out.println(num2);
        if(op == '+'){
            System.out.println(num1+num2);
        }
    }
}
```

```

    }
    else if(op == '-'){
        System.out.println(num1-num2);
    }
}
}
}

```

RUBRIC

SL	Points to Meet	Marks [10]
1	Importing Scanner class, main method, and Java syntax	0.5
2	Correctly takes two strings and an operator from the user	1
3	Properly handles the iteration and extracts the characters of both Strings	1.5
4	Checking the digits using characters ('0'-'9') or ASCII values [both Strings]	1
5	Converting the character to a digit for both Strings	1
6	Searching the even digits [both Strings]	0.5
7	Initialize num1, num2, and correctly prepare the numbers	3
8	Checking the operations (+,-) and performing the correct operation	1
9	Printing all three outputs correctly	0.5

Name:	ID:	Section:
-------	-----	----------

Question 1 [C03] [10 Points]

Write a Java program that rearranges the elements of a given array in such a way that all even numbers appear before all odd numbers. *[Do it in-place. You can't create a new array to solve it. The relative order among even or odd numbers does not matter.] [Your code should work for any given integer array]*

Given Array	Expected Output
<code>int [] arr = { 3, 6, 4, 7, 1, 9, 2};</code>	Rearranged Array: 6, 4, 2, 7, 1, 9, 3
<code>int[] arr = { 2, 11, 1, 7, 12, 8, 3};</code>	Rearranged Array: 2, 12, 8, 7, 11, 1, 3

Name:	ID:	Section:
-------	-----	----------

Question 1 [C03] [10 Points]

Write a Java program that rearranges the elements of a given array in such a way that all odd numbers appear before all even numbers. *[Do it in-place. You can't create a new array to solve it. The relative order among even or odd numbers does not matter.] [Your code should work for any given integer array]*

Given Array	Expected Output
<code>int [] arr = {5, 4, 2, 7, 8, 9};</code>	Rearranged Array: 5, 7, 9, 4, 8, 2
<code>int[] arr = { 4, 16, 14, 3, 7, 12, 5};</code>	Rearranged Array: 3, 7, 5, 4, 16, 12, 14

In case of any circumstance not present in the rubrics, feel free to discuss in the assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

SET - [A]

SOLUTION

```
public class EvenOddInPlace {
    public static void main(String[] args) {
        int[] arr = { 3, 6, 4, 7, 1, 9, 2};
        int pos = 0;
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] % 2 == 0) {
                int temp = arr[i];
                arr[i] = arr[pos];
                arr[pos] = temp;
                pos++;
            }
        }
        System.out.println("Rearranged Array:");
        for (int i = 0; i < arr.length; i++) {
            if(i==arr.length-1){
                System.out.println(arr[i]);
            }
            else{
                System.out.print(arr[i] + ", ");
            }
        }
    }
}
```

Alternate Solution

```
public class EvenOddNested {
    public static void main(String[] args) {
        int[] arr = {3, 6, 4, 7, 1, 9, 2};
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] % 2 != 0) {
                for (int j = i + 1; j < arr.length; j++) {
                    if (arr[j] % 2 == 0) {
                        int temp = arr[i];
                        arr[i] = arr[j];
                        arr[j] = temp;
                        break;
                    }
                }
            }
        }
        System.out.println("Rearranged Array:");
        for (int i = 0; i < arr.length; i++) {
            if (i == arr.length - 1) {
                System.out.println(arr[i]);
            } else {
                System.out.print(arr[i] + ", ");
            }
        }
    }
}
```

RUBRIC		
SL	Points to Meet	Marks [10]
1	Main method, and basic Java syntax	0.5
2	Correct array declaration and initialization of required variables	1
3	Loop for traversing the array	1.25
4	Checking even numbers using condition (% 2 == 0)	1.5
5	Swapping logic (in-place rearrangement)	2
6	Updating variables correctly	1.5
7	Loop for printing array elements	1.25
8	Proper output formatting (handling comma at the end)	1

SET - [B]

SOLUTION
<pre> public class EvenOddInPlace { public static void main(String[] args) { int[] arr = { 3, 6, 4, 7, 1, 9, 2}; int pos = 0; for (int i = 0; i < arr.length; i++) { if (arr[i] % 2 != 0) { int temp = arr[i]; arr[i] = arr[pos]; arr[pos] = temp; pos++; } } System.out.println("Rearranged Array:"); for (int i = 0; i < arr.length; i++) { if(i==arr.length-1){ System.out.println(arr[i]); } else{ System.out.print(arr[i] + ", "); } } } } </pre>

Alternate Solution

```
public class EvenOddNested {
    public static void main(String[] args) {
        int[] arr = {3, 6, 4, 7, 1, 9, 2};
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] % 2 == 0) {
                for (int j = i + 1; j < arr.length; j++) {
                    if (arr[j] % 2 != 0) {
                        int temp = arr[i];
                        arr[i] = arr[j];
                        arr[j] = temp;
                        break;
                    }
                }
            }
        }

        System.out.println("Rearranged Array:");
        for (int i = 0; i < arr.length; i++) {
            if (i == arr.length - 1) {
                System.out.println(arr[i]);
            } else {
                System.out.print(arr[i] + ", ");
            }
        }
    }
}
```

RUBRIC

SL	Points to Meet	Marks [10]
1	Main method, and basic Java syntax	0.5
2	Correct array declaration and initialization of required variables	1
3	Loop for traversing the array	1.25
4	Checking odd numbers using condition (% 2 != 0)	1.5
5	Swapping logic (in-place rearrangement)	2
6	Updating variables correctly	1.5
7	Loop for printing array elements	1.25
8	Proper output formatting (handling comma at the end)	1